Loan Default Prediction System: Explainable Credit Risk Scoring with Drift Monitoring

HCLTech Career Shaper Programme | III Year Industry Project | Project No. 137

Source code and deployment

Source code (GitHub)	[GITHUB REPOSITORY URL]
Run locally	FastAPI + uvicorn on http://127.0.0.1:8000 (Swagger UI at /docs)
Hosting	Local Windows machine; no cloud deployment (out of scope)
Model and API	Python 3.13, scikit-learn 1.8, XGBoost 3.2, SHAP 0.51, FastAPI 0.141, Pydantic 2.13, SQLite
Data and analysis	pandas 3.0, matplotlib, seaborn, Evidently 0.7, Jupyter
Server / database	uvicorn ASGI server; SQLite file logs/predictions.db

Run the API

Requires Python 3.11 to 3.14 (3.13 recommended) on Windows. Save the Kaggle CSV as data\raw\credit_risk_dataset.csv, then start the service and open http://127.0.0.1:8000/docs:

```
py -3.13 -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
.\.venv\Scripts\python.exe -m uvicorn src.api.main:app --port 8000
```

Rerun the data and model pipeline

From the project folder, in this order (src.robustness takes about 1 minute):

```
.\.venv\Scripts\python.exe -m src.data_prep     # clean + 70/15/15 split
.\.venv\Scripts\python.exe -m src.train         # 3 models, best on validation AUC
.\.venv\Scripts\python.exe -m src.evaluate      # test metrics, bands, NPA back-test
.\.venv\Scripts\python.exe -m src.robustness    # 5-fold CV + ablation
.\.venv\Scripts\python.exe -m src.explain       # SHAP importance + reasons
.\.venv\Scripts\python.exe -m src.drift         # PSI table + Evidently HTML report
.\.venv\Scripts\python.exe -m pytest -q tests   # 13 passed
```

Repository layout

```
loan-default-prediction/
├── src/          pipeline: data_prep, train, evaluate, robustness, explain, drift
│   └── api/      FastAPI service: schemas.py, store.py (SQLite log), main.py
├── scripts/      simulate_traffic.py: sends normal and downturn traffic to the API
├── tests/        13 pytest tests for the data, the model and the API
├── notebooks/    01_eda.ipynb, the executed EDA notebook
├── data/         raw Kaggle CSV and the cleaned train/val/test split (git-ignored)
├── models/       model_v1.joblib and metadata_v1.json (candidates/ is git-ignored)
├── reports/      evaluation, robustness, SHAP and drift results; charts in eda/
├── logs/         predictions.db, written by the API (git-ignored)
├── _evidence/    console output of every pipeline, API and test run
├── submission/   HCL documents, figures and the scripts that build them
└── BUILD_LOG.md  README.md  requirements.txt
```